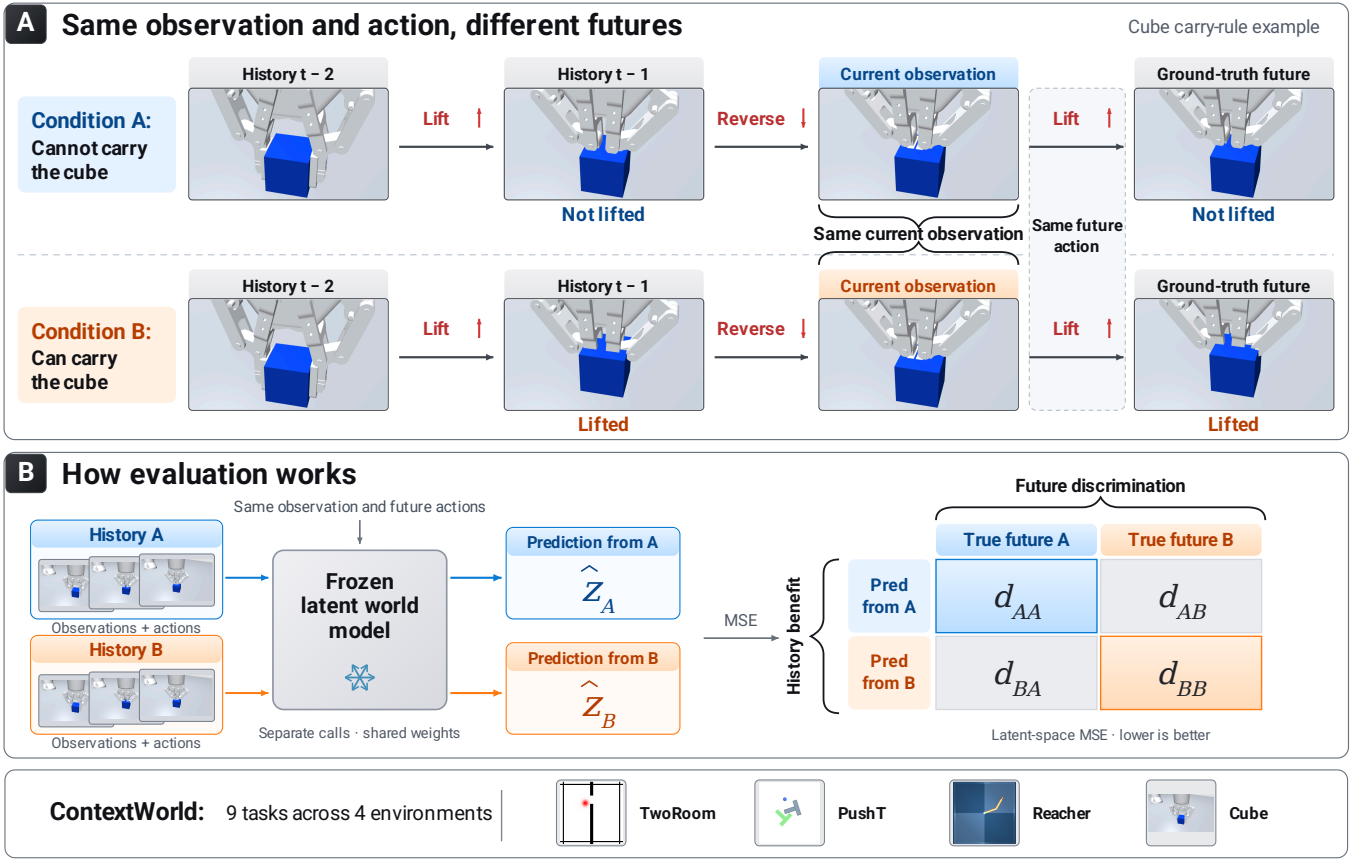

A
Same observation and action, different futures
Cube carry-rule example
Condition A:
Cannot carry the cube
History t − 2
Lift
History t − 1
Not lifted
Reverse
Current observation
Same current observation
Lift
Same future action
Ground-truth future
Not lifted
Condition B:
Can carry the cube
History t − 2
Lift
History t − 1
Lifted
Reverse
Current observation
Lift
Ground-truth future
Lifted
B
How evaluation works
Same observation and future actions
History A
Observations + actions
History B
Observations + actions
Frozen latent world model
Separate calls · shared weights
Prediction from A
$\hat{z}_A$
Prediction from B
$\hat{z}_B$
MSE
Future discrimination
True future A
True future B
History benefit
Pred from A
Pred from B
d_{AA}
d_{AB}
d_{BA}
d_{BB}
Latent-space MSE · lower is better
ContextWorld:
9 tasks across 4 environments
TwoRoom
PushT
Reacher
Cube